

ISM Note 13

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1. Turbulence (continued)

Kolmogorov Power Spectrum

Recall that we have derived the Kolmogorov power spectrum last time, which is given by

$$E(k) \propto k^{-5/3}$$

Since $k \sim \frac{1}{L}$, we can see that large eddies carry more energy than small eddies.

The lower limit of the inertial range is called **the injection scale**, which corresponds to the largest eddy. Within the inertial range, the energy transfer is called the **turbulence cascade**. The upper limit of the inertial range is called the **dissipation scale**, which corresponds to the smallest eddy. Over the dissipation scale, the kinetic energy is converted into thermal energy.

We need to point out that the cascade occurs in 3 dimensions, while in 2 dimension the cascade is inversed, and we call it the inverse cascade.

MHD Turbulence

The most famous theory for the MHD turbulence is the so-called Goldreich-Sridhar 1995 theory(GS95). In this setup, they still assume the turbulence is incompressible, but now the magnetic strength is comparably strong.

We can decompose the velocity into two components, which are the parallel component and the perpendicular component. The critical balance condition is given by

$$\frac{L_{\perp}}{\delta u} \sim \frac{L_{\parallel}}{v_A}$$

The LHS is the non-linear distortion timescale, and the RHS is the Alfven timescale. And since we know from the Kolmogorov power spectrum that

$$\delta u \propto L_{\perp}^{1/3}$$

We can get a power law

$$L_{\parallel} \propto L_{\perp}^{2/3}$$

So what does this physically mean? The thing is, if you go to the small scale, e.g. $k_{\perp}$ is large, $k_{\parallel}$ does not grow as fast as $k_{\perp}$. Therefore, the parallel component is much smaller than the perpendicular component, which means the longer perturbation structure exist along B direction. **So MHD turbulence is anisotropic and elongated along B field lines, especially at small scales.**

We can understand it in another way. In the large scale, there is no time for the Alfven wave to propagate and different two points are disconnected, so you will not see strong MHD effect.

The Origin of Astrophysical Turbulence

The key thing here is, you have multiple scales of injection. For example,

Mpc scale

- the cosmological inflow,
- virial shock,
- the merge of galaxy cluster

kpc scale

- the galaxy related gravitational instability,
- galaxy merger,
- disk instability

pc scale

- the galactic feedback, e.g. supernova feedback, star formation feedback, AGN feedback
- stellar winds,
- supernova remnants,
- some ISM thermal instabilities

sub-pc scale

- the fluid instabilities,
- plasma instabilities,
- cosmic rays

Consequence of Turbulence

First of all, turbulence can create density fluctuations, this can then **lead to thermal instability, gravitational instability** etc.

The second part is the so-called **turbulent mixing**. The turbulence can mix both the gas phases and the metal in a very efficient way.

Also, we know that the turbulence can **provide pressure support**, which is

$$P_{\text{turb}} \sim \rho v_{\text{turb}}^2$$

And then, the turbulence is an important source of **heating**, which will convert the kinetic energy into thermal energy.

What's more, the turbulent dynamo can **amplify the magnetic fields**.

The turbulence can also accounts for **cosmic ray acceleration and transport**.

Complexity Beyond Kolmogorov

In astrophysical environment, the turbulence can be inhomogeneous in both space and time. The spatial inhomogeneity is called the **intermittency**.

And also, the turbulence can be anisotropic, like the MHD turbulence we have discussed and the turbulence in a gravitationally stratified medium.

Besides, the turbulence can be compressible, non-adiabatic and so on. In all these cases, the Kolmogorov power spectrum will not hold correct.

How to Observe Turbulence in ISM/CGM/ICM ...

The first thing is, we can measure them from **absorption**. Consider a light transmit through a medium, there will be some ions absorbing the light, and there will be a **broadening of the absorption line**. The broadening can be given by

$$b_{obs} = \frac{2k_B T}{m} + b_{nt}$$

where the first term is the thermal broadening, and the second term is the non-thermal broadening, which can be caused by the turbulence. Measuring the non-thermal broadening can tell us the turbulence velocity **parallel** to the line of sight.

The second thing of course is the **emission**. We can measure the width of the emission line to observe the turbulence, which can tell us the turbulence velocity **perpendicular** to the line of sight. However, in the diffuse gas we usually care about, the emission is very weak, unlike the absorption where we can use the bright background source like a quasar.

2. Cosmic Rays

The first question is, what are cosmic rays? Cosmic rays are highly relativistic particles, which can be protons, electrons, positrons etc.

The distribution function of cosmic rays is nearly a piecewise power law, which is shown in the figure below.

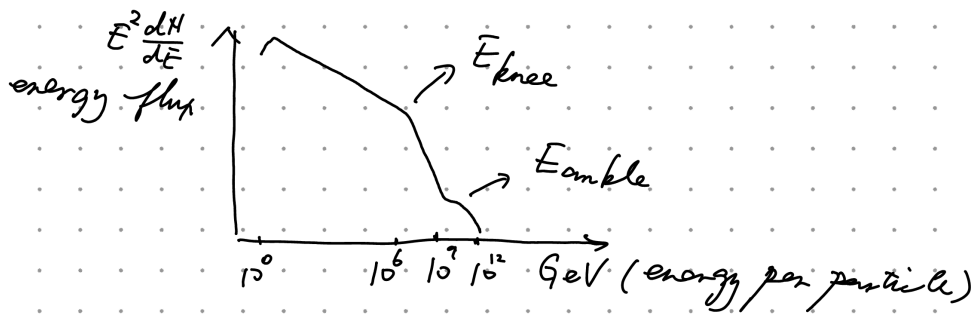


Fig 1. Cosmic ray spectrum.

This distribution function is non-Maxwellian and non-thermal. The two part of power law has different acceleration mechanisms.

For galaxy evolution, we mostly care about GeV proton cosmic rays, that is because GeV cosmic rays have largest flux, and protons have much longer lifetime than electrons.

The cosmic rays are quite collisionless. That's because the mean free path of cosmic rays is very large, which is usually $\lambda_{\text{mfp}} \sim 10^{30}$ cm for GeV cosmic rays.

The cosmic rays have extremely low level of anisotropy, which is about $\delta \sim 10^{-4}$, so the drift bulk motion is about $v_{\text{bulk}} \sim 10^{-4}c \ll c$. Then the escape time from galaxy can be estimated as

$$t_{\text{esc}} \sim \frac{H}{v_{\text{bulk}}} \sim 5 \text{ Myr}$$

Therefore, the cosmic rays can remain in the galaxy for a long time and the feedback of CR can be important.

Gyro Motion

The gyro radius of cosmic rays is given by

$$r_g \sim \frac{p_{\perp} c}{qB} \sim \frac{2.2}{Z} \left(\frac{p_{\perp}}{10 \text{ GeV}/c} \right) \left(\frac{B}{1 \mu\text{G}} \right) \text{ AU}$$

The gyro radius is much smaller than the relevant galactic length, so the cosmic rays can stay in the galaxy for a long time.

More details will be discussed in the next lecture.

